

Data on Size of the Brain and of Various Brain Parts in Insectivores and Primates

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Two papers of this volume are based on data acquired and collected by Stephan, Bauchot, and Andy since 1956 about the size of the brain and of various brain parts. One of these papers (Stephan and Andy, Chapter 5) deals with aspects related to evolution and function of the allocortex; the other (Sacher, Chapter 9) is an extensive factorial analysis of these data.

Their publication, besides presenting the reader with an opportunity to evaluate critically the results of these papers, makes possible the comparison with similar data obtained in other laboratories, allows further analysis with other appropriate mathematical methods, and permits the comparison with behavioral observations.

Since Primates have their phylogenetic origin in insectivore-like ancestors, the data on volume measurements in insectivore brains is also given. We have identified the species with the most primitive cerebral pattern and grouped them together as *Basal Insectivores* (Table 1). These are expected to show the least degree of change since their first appearance (permanent types) and they may still be comparatively similar to the early forerunners of Primates. Thus, they constitute a good base of reference for evaluating evolutionary progress. In contrast, the remainder of the Insectivores reveal distinct marks of higher development, and as such, have been termed *Progressive Insectivores* (Table 2).

The volume data have been determined from serial sections from a total of 63 species in our collection, i.e., from 22 Insectivores, 20 Prosimians and 21 Simians (Tables 1-6). The volumes were estimated from enlarged photographs of serial sections taken at equal intervals (60-80 per brain). The sections were projected

TABLE 1

Data on body and brain weight (in g and mg) and volume of the fundamental brain sections (in mm³) in Insectivores.

	Body Weight * (g)	Brain Weight † (mg)	Total Brain ‡ (net volume)	Medulla oblongata §	Cerebellum	Mesencephalon	Diencephalon	Telencephalon
	1	2	3	4	5	6	7	8
A. Basal Insectivores								
1. <i>Sorex minutus</i>	5.3	110	103	17.0	11.0	5.9	8.7	60.7
2. <i>Sorex araneus</i>	10.3	200	188	26.3	18.3	10.6	15.9	117
3. <i>Crocidura russula</i>	11.0	190	178	23.6	21.8	11.7	15.0	106
4. <i>Crocidura occident.</i>	28.0	440	408	59.8	51.3	27.7	28.4	241
5. <i>Suncus murinus</i>	35.0	380	354	50.5	44.8	24.2	26.0	209
6. <i>Echinops telfairi</i>	87.5	620	569	102	56.0	45.5	40.0	326
7. <i>Hemicentetes semisp.</i>	110	830	757	108	117	44.8	48.5	438
8. <i>Setifer setosus</i>	248	1,510	1,404	183	142	75.7	99.7	903
9. <i>Tenrec ecaudatus</i>	832	2,570	2,336	309	339	137	169	1,381
10. <i>Erinaceus europaeus</i>	860	3,350	2,969	289	388	142	229	1,921
11. <i>Aethechinus algirus</i>	700	3,550	3,174	397	407	157	285	1,927
B. Progressive Insectivores								
12. <i>Solonodon paradoxus</i>	900	4,670	4,262	493	807	204	272	2,486
13. <i>Nesogale talazaci</i>	50.4	790	741	85.3	92.3	47.2	43.8	472
14. <i>Limnogale mergulus</i>	92.0	1,150	1,046	204	176	67.4	81.0	518
15. <i>Potamogale velox</i>	660	4,100	3,822	746	597	250	294	1,936
16. <i>Neomys fodiens</i>	15.2	320	299	42.2	38.9	17.3	25.2	175
17. <i>Talpa europaea</i>	76.0	1,020	953	95.5	174	49.8	80.2	554
18. <i>Galemys pyrenaicus</i>	57.5	1,330	1,230	147	184	62.2	105	733
19. <i>Desmana moschata</i>	440	4,000	3,620	402	514	219	344	2,140
20. <i>Chorotalpa stuhlmanni</i>	39.8	740	693	68.3	74.7	42.0	51.8	457
21. <i>Elephantulus fuscipes</i>	57.0	1,330	1,233	120	183	117	112	700
22. <i>Rhynchocyon stuhlmanni</i>	490	6,100	5,680	507	798	523	524	3,328

* Body weight from Bauchot and Stephan (1966, 1969) with the exception of the values of *Nycticebus cougang* and *Macaca mulatta*.

† Brain weight from Bauchot and Stephan (1966, 1969) with exception of the value of *Macaca mulatta*.

‡ Total brain (net volume). All volume data are net values, that is they give the volumes of the pure tissues: ventricles, meningeal membranes and nerves are excluded.

§ Medulla oblongata. Since it is almost impossible to separate the Substantia reticularis of the Medulla oblongata from that of the Mesencephalon we have included the latter into the Medulla oblongata.

|| Cerebellum. Included are Brachium and Nuclei pontis.

directly on photographic paper, the borders delineated, and the components of each structure then cut out and weighed. The photographic paper was previously weighed in order to determine the number of square centimeters per gram and care was taken to compensate for changes in paper weight due to varying temperature and humidity. Distances between the histologic sections were calculated, and finally, volumes for each division were derived by applying the formula:

$$V = \frac{AP \times WS \times D}{M^2}$$

TABLE 2.

Data on body and brain weight (in g and mg) and volume of the fundamental brain sections (in mm³) in Prosimians.

	Body Weight * (g)	Brain Weight † (mg)	Total Brain ‡ (net volume)	Medulla oblongata §	Cerebellum	Mesencephalon	Diencephalon	Telencephalon
	1	2	3	4	5	6	7	8
23. <i>Tupaia glis</i>	150	3,150	2,959	230	406	261	285	1,778
24. <i>Urogale everetti</i>	275	4,280	3,997	355	513	307	394	2,428
25. <i>Microcebus murinus</i>	54.0	1,780	1,663	82.6	226	70.1	127	1,157
26. <i>Cheirogaleus medius</i>	177	3,140	2,941	206	401	144	245	1,945
27. <i>Cheirogaleus major</i>	450	6,800	6,323	370	938	224	482	4,309
28. <i>Lepilemur ruficaud.</i>	915	7,600	7,167	433	1,107	283	614	4,729
29. <i>Hapalemur simus</i>	1,300	9,530	8,868	485	1,419	313	777	5,875
30. <i>Lemur fulvus</i>	1,400	23,300	22,053	802	3,411	536	1,580	15,724
31. <i>Lemur variegatus</i>	3,000	31,500	29,713	1,420	4,286	1,018	2,529	20,461
32. <i>Avahi laniger</i>	860	9,670	9,075	549	1,423	345	875	5,882
33. <i>Propithecus verr.</i>	3,480	26,700	25,080	1,199	3,923	741	2,071	17,146
34. <i>Indri indri</i>	6,250	38,300	36,159	1,376	5,441	972	3,101	25,269
35. <i>Daubentonia</i> <i>madagasc.</i>	2,800	45,150	42,611	1,517	6,461	897	3,539	30,196
36. <i>Loris gracilis</i>	322	6,600	6,269	233	728	220	538	4,551
37. <i>Nycticebus cougang</i>	800	12,500	11,755	528	1,310	345	1,077	8,495
38. <i>Perodicticus potto</i>	1,150	14,000	13,212	680	1,699	391	1,024	9,418
39. <i>Galago demidovii</i>	81.0	3,380	3,203	169	413	135	246	2,240
40. <i>Galago senegalensis</i>	186	4,800	4,512	254	672	205	382	2,997
41. <i>Galago</i> <i>crassicaudatus</i>	850	10,300	9,602	477	1,361	350	748	6,666
42. <i>Tarsius syrichta</i>	87.5	3,630	3,416	185	422	171	279	2,359

* Body weight from Bauchot and Stephan (1966, 1969) with the exception of the values of *Nycticebus cougang* and *Macaca mulatta*.

† Brain weight from Bauchot and Stephan (1966, 1969) with exception of the value of *Macaca mulatta*.

‡ Total brain (net volume). All volume data are net values, that is they give the volumes of the *pure tissues*: ventricles, meningeal membranes and nerves are excluded.

§ Medulla oblongata. Since it is almost impossible to separate the Substantia reticularis of the Medulla oblongata from that of the Mesencephalon we have included the latter into the Medulla oblongata.

|| Cerebellum. Included are Brachium and Nuclei pontis.

V = volume in mm³

AP = average area of the photographic paper in mm² per mg

WS = Weight of the photographic paper of the cut out structure in milligrams

D = distance of measured sections in mm

M² = square of linear magnification

The volume of the total brain obtained in this manner is markedly smaller than the volume of the fresh brain. This is due to the shrinkage which results from the fixation and embedding procedures.

The degree of shrinkage is different for each brain, even if these procedures are

TABLE 3.

Data on body and brain weight (in g and mg) and volume of the fundamental brain sections (in mm³) in Simians.

	Body Weight *	Brain Weight †	Total Brain ‡	Medulla §	Cerebellum	Mesencephalon	Diencephalon	Telencephalon
	(g)	(mg)	(net volume)	oblongata §				
	1	2	3	4	5	6	7	8
43. <i>Callithrix jacchus</i>	260	7,600	7,248	334	754	295	549	5,316
44. <i>Saguinus oedipus</i>	405	10,000	10,576	345	958	340	729	7,203
45. <i>Saguinus tamarin</i>	340	10,300	9,459	434	962	319	706	7,038
46. <i>Aotes trivirgatus</i>	850	16,000	15,229	641	1,661	427	1,086	11,414
47. <i>Callicebus moloch</i>	650	15,500	14,434	683	1,287	469	1,101	10,894
48. <i>Pithecia monacha</i>	1,500	35,000	32,836	1,026	3,961	782	2,167	24,901
49. <i>Alouatta seniculus</i>	6,400	51,000	47,749	1,840	6,349	1,039	3,217	35,303
50. <i>Cebus sp.</i>	2,600	73,000	68,672	1,844	8,560	1,153	3,999	53,116
51. <i>Cebus albifrons</i>	3,000	80,000	75,592	1,896	8,357	1,487	4,623	59,229
52. <i>Saimiri sciureus</i>	680	22,000	20,691	661	2,071	482	1,311	16,165
53. <i>Ateles geoffroyi</i>	8,000	108,000	101,034	1,834	12,438	1,482	5,334	79,946
54. <i>Lagothrix logotricha</i>	5,200	101,000	94,939	1,742	12,016	1,423	5,206	74,553
55. <i>Macaca mulatta</i>	6,000	93,000	87,896	1,992	8,965	1,380	4,480	71,080
56. <i>Cercocebus albigena</i>	7,900	104,000	97,603	2,708	10,725	1,770	5,351	77,049
57. <i>Cercopithecus talapoin</i>	1,000	39,000	36,830	1,139	3,085	767	2,460	29,379
58. <i>Cercopithecus ascanius</i>	3,600	65,000	61,610	1,583	5,654	1,127	3,498	49,748
59. <i>Cercopithecus mitis</i>	6,500	76,000	71,505	2,026	6,848	1,372	4,232	57,027
60. <i>Colobus badius</i>	7,000	78,000	73,818	2,007	8,648	1,333	3,945	57,885
61. <i>Pan troglodytes</i>	46,000	420,000	396,255	6,032	45,280	3,878	15,962	325,104
62. <i>Gorilla gorilla</i>	125,000	465,000	437,433	6,983	64,402	4,047	18,014	343,987
63. <i>Homo sapiens</i>	65,000	1,330,000	1,251,847	9,622	137,421	8,087	33,319	1,063,398

* Body weight from Bauchot and Stephan (1966, 1969) with the exception of the values of *Nycticebus cougang* and *Macaca mulatta*.

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‡ Total brain (net volume). All volume data are net values, that is they give the volumes of the *pure tissues*: ventricles, meningeal membranes and nerves are excluded.

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|| Cerebellum. Included are Brachium and Nuclei pontis.

the same. In order to obtain comparable values it is necessary to correct the figures to the volume of the *fresh* brain. The latter can be found by dividing the weight of the fresh brain by the specific brain weight.

The specific gravity of fresh brains were found to be close to 1.036 grams/cm³ (Stephan, 1960). Thus:

$$\text{Volume of fresh brain} = \frac{\text{weight of fresh brain}}{1.036}$$

To obtain the factor of correction (FC) this volume must be divided by the serial section volume:

$$\text{FC} = \frac{\text{volume of fresh brain}}{\text{serial section volume}}$$

TABLE 4.

Volume of the various components of the telencephalon in Insectivores (in mm³).

	Bulbus ol- factorius 9	Paleocortex + NA* 10	Septum† 11	Striatum‡ 12	Schizocortex§ 13	Hippocampus 14	Neocortex 15
A. Basal Insectivores							
1. Sorex minutus	8.7	17.2	2.0	6.0	3.6	10.5	12.7
2. Sorex araneus	14.2	32.7	3.8	10.1	7.2	20.6	28.1
3. Crocidura russula	16.4	30.2	3.6	8.2	6.1	15.9	25.5
4. Crocidura occident.	37.8	68.4	6.9	16.8	13.3	38.5	59.1
5. Suncus murinus	34.4	56.2	5.5	14.8	12.6	31.2	54.1
6. Echinops telfairi	65.7	118	11.0	24.1	9.6	49.6	48.5
7. Hemientetes semisp.	93.8	148	12.8	29.8	17.4	63.7	73.4
8. Setifer setosus	210	338	26.0	55.6	41.9	97.7	134
9. Tenrec ecaudatus	293	452	37.9	101	71.3	146	280
10. Erinaceus europaeus	350	507	46.2	134	120	220	543
11. Aethechinus algirus	297	504	41.6	163	113	268	541
B. Progressive Insectivores							
12. Solonodon paradoxus	478	683	68.1	235	140	299	583
13. Nesogale talazaci	74.6	139	11.2	31.2	20.7	98.9	96.2
14. Limnogale mergulus	43.2	87.4	19.5	57.9	36.4	78.2	196
15. Potamogale velox	87.0	191	50.7	132	142	294	1,039
16. Neomys fodiens	15.9	36.0	6.4	18.7	9.0	32.9	56.6
17. Talpa europaea	60.0	112	14.9	60.2	39.6	86.7	181
18. Galemys pyrenaicus	39.2	107	20.1	74.4	44.9	109	339
19. Desmana moschata	142	320	42.6	222	163	267	983
20. Chorotalpa stuhlmanni	60.8	131	11.3	43.5	21.4	59.1	129
21. Elephantulus fuscipes	63.9	121	16.8	44.5	46.8	171	236
22. Rhynchocyon stuhlmanni	427	705	81.5	222	220	582	1,089

* Paleocortex + NA. The Corpus amygdaloideum (Nucleus amygdalae, NA) has so far not been separated from the Palaeocortex since it is difficult to delineate the periamygdaloid cortex from the subcortical nuclei. A more detailed study is in progress.

† Septum. Included is the Diagonal band of Broca in its whole extent.

‡ Striatum. Included is the internal capsule since it is strongly dispersed in the low insectivores and therefore difficult to separate from striatum.

§ Schizocortex. It is made up of the Regio entorhinalis, Regio perirhinalis and Regiones prae- and parasubicularis, and characterized by the presence of one or several, almost cell-free zones between the various layers (Rose, 1927).

|| Neocortex. Includes the underlying white substance.

The size of this factor of correction is clearly distinct in different types of fixation. The following means were found:

FIXATION FLUID	N	Factor of correction		Volume loss in % of fresh brain	
		MEAN	RANGE	MEAN	RANGE
Bouin's fluid	45	1.97 ± 0.16	1.72 — 2.23	49.2	41.9 — 55.2
Formol-alcohol	2	1.47 ± 0.11	1.37 — 1.58	32.2	27.0 — 36.7
Formol 10%	8	2.43 ± 0.17	2.26 — 2.74	58.8	55.8 — 63.5

TABLE 5.

Volume of the various components of the telencephalon in Prosimians (in mm³).

	Bulbus ol- factorius 9	Palaeocortex + NA* 10	Septum† 11	Striatum‡ 12	Schizocortex§ 13	Hippocampus 14	Neocortex 15
23. <i>Tupaia glis</i>	128	256	32.1	133	84.2	138	1,007
24. <i>Urogale everetti</i>	186	362	45.1	199	124	173	1,338
25. <i>Microcebus murinus</i>	40.3	94.5	15.5	94.3	39.4	100	773
26. <i>Cheirogaleus medius</i>	99.3	181	31.5	146	59.5	175	1,252
27. <i>Cheirogaleus major</i>	155	315	64.1	336	144	332	2,963
28. <i>Lepilemur ruficaud.</i>	131	287	57.1	388	199	380	3,287
29. <i>Hapalemur simus</i>	79.4	223	62.1	441	214	525	4,331
30. <i>Lemur fulvus</i>	229	638	121	1,152	433	709	12,442
31. <i>Lemur variegatus</i>	374	985	206	1,591	608	1,404	15,293
32. <i>Avahi laniger</i>	80.6	256	78.1	444	206	476	4,341
33. <i>Propithecus verr.</i>	168	565	160	1,425	395	1,044	13,389
34. <i>Indri indri</i>	142	830	223	1,755	808	1,535	19,976
35. <i>Daubentonia madagasc.</i>	693	1,538	254	2,765	1,044	1,776	22,127
36. <i>Loris gracilis</i>	88.1	256	41.1	351	99.7	191	3,524
37. <i>Nycticebus cougang</i>	164	510	92.0	760	212	566	6,192
38. <i>Perodicticus potto</i>	312	661	114	712	328	607	6,683
39. <i>Galago demidovii</i>	84.4	181	25.7	169	59.9	152	1,568
40. <i>Galago senegalensis</i>	81.8	163	35.7	210	107	261	2,139
41. <i>Galago crassicaudatus</i>	180	405	70.6	504	178	412	4,919
42. <i>Tarsius syrichta</i>	18.1	95.0	20.9	133	64.3	138	1,890

* Palaeocortex + NA. The Corpus amygdaloideum (Nucleus amygdalae, NA) has so far not been separated from the Palaeocortex since it is difficult to delineate the periamygdaloid cortex from the subcortical nuclei. A more detailed study is in progress.

† Septum. Included is the Diagonal band of Broca in its whole extend.

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§ Schizocortex. It is made up of the Regio entorhinalis, Regio perirhinalis and Regiones prae- and parasubicularis, and characterized by the presence of one or several, almost cell-free zones between the various layers (Rose, 1927).

|| Neocortex. Includes the underlying white substance.

Because of these large differences it is impossible to compare uncorrected figures from material with unknown or different types of fixation. Even if the same fixation and embedding procedures are applied to a series of brains, corrections remain appropriate because of the wide range of volume loss to be expected as indicated in the last column of the above table.

This factor of correction calculated for the *total* brain is then used to correct all the *various parts*. This may not be quite correct, since various parts of the brain may have somewhat different degrees of shrinkage. However, such differences are difficult to ascertain and since they are expected to be small, they were not taken into account.

We have endeavored to include representatives from as many species as possible. But the majority of the species here investigated are rare and/or valuable; and the

TABLE 6.

*Volume of the various components of the telencephalon in Simians
(in mm³).*

	Bulbus ol- factorius 9	Paleocortex + NA* 10	Septum† 11	Striatum‡ 12	Schizocortex§ 13	Hippocampus 14	Neocortex 15
43. <i>Callithrix jacchus</i>	28.4	224	49.7	332	85.2	238	4,357
44. <i>Saguinus oedipus</i>	24.6	282	68.6	420	99.6	259	6,050
45. <i>Saguinus tamarin</i>	16.8	203	43.6	415	113	278	5,969
46. <i>Aotes trivirgatus</i>	60.3	394	92.9	779	255	479	9,354
47. <i>Callicebus moloch</i>	16.8	425	64.7	721	212	481	8,973
48. <i>Pithecia monacha</i>	38.2	699	123	1,642	283	850	21,266
49. <i>Alouatta seniculus</i>	45.2	943	154	2,566	428	1,325	29,843
50. <i>Cebus</i> sp.	49.6	1,186	212	3,295	436	892	47,045
51. <i>Cebus albifrons</i>	37.2	892	161	3,635	401	1,029	53,073
52. <i>Saimiri sciureus</i>	25.4	411	83.0	929	154	316	14,246
53. <i>Ateles geoffroyi</i>	92.6	1,625	324	4,950	732	1,366	70,856
54. <i>Lagothrix logotricha</i>	83.0	1,502	241	4,662	597	1,381	66,087
55. <i>Macaca mulatta</i>	84.3	1,220	271	4,032	639	1,353	63,481
56. <i>Cercocebus albigena</i>	121	1,639	294	4,146	630	1,485	68,733
57. <i>Cercopithecus talapoin</i>	22.3	732	121	1,835	237	684	25,747
58. <i>Cercopithecus ascanius</i>	96.5	1,136	243	2,627	673	1,154	43,818
59. <i>Cercopithecus mitis</i>	118	1,282	249	2,769	625	1,385	50,599
60. <i>Colobus badius</i>	51.3	998	288	3,156	814	1,671	50,906
61. <i>Pan troglodytes</i>	267	2,852	882	12,700	2,093	3,919	302,392
62. <i>Gorilla gorilla</i>	294	4,530	1,090	13,547	2,537	4,446	317,542
63. <i>Homo sapiens</i>	114	9,161	2,610	28,559	6,142	10,287	1,006,525

* Paleocortex + NA. The Corpus amygdaloideum (Nucleus amygdalae, NA) has so far not been separated from the Palaeocortex since it is difficult to delineate the periamygdaloid cortex from the subcortical nuclei. A more detailed study is in progress.

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|| Neocortex. Includes the underlying white substance.

type of quantitative investigation presented here is extremely time consuming. It was therefore necessary to restrict ourselves to a very limited number of specimens.

Nevertheless, to obtain volumes which are as representative as possible we have performed a second correction. This was accomplished by taking into account the difference between the individual weight of the brain treated here and the standard brain weight of each species as given by Bauchot and Stephan (1966, 1969) from more comprehensive material. Such differences are due to intraspecific variations in brain weights.

$$C = \frac{\text{weight of standard fresh brain}}{\text{weight of individual fresh brain}}$$

This second correction (C) once again was determined by comparing *total* brains, and utilized to correct all the various parts of the brain. This is possible, since according to our experience, extremely small or extremely large brains within one species have generally the same composition. There is no indication of directed changes in composition related to the size of the brain, only a normal range of variation.

In most species, only one brain was investigated. However, in order to get some indication of the degree of intraspecific variation, we investigated two brains of 18 species (table numbers 2, 3, 4, 8, 10, 15, 16, 17, 18, 20, 21, 22, 23, 37, 38, 39, 57, 60). The *mean* of the two individual values for each structure is given in the tables and the percentage deviation of the individual values from these means has been calculated. The average of the total of 216 values (12 structures from 18 species, respectively) is 3.7 ranging from 0 to ± 21.3 . Eighty per cent of all values fall below 6 and 95% below 10. Thus we expect, that in future investigations of a more extensive material the values given here will not vary more than 10%. Similar results were reported also for Chiroptera (Stephan and Pirlot, 1970). More than 10% deviation of the individual values from the mean were found in the striatum of *Erinaceus europaeus* (11.5), mesencephalon and olfactory bulb of *Neomys fodiens* (10.5 and 13.1), neocortex of *Talpa europaea* (21.3), schizocortex of *Chlorotalpa stuhlmanni* (15.1), olfactory bulb and palaeocortex of *Perodicticus potto* (11.3 and 15.1), schizocortex by *Nycticebus cougang* (12.4) and medulla oblongata and olfactory bulb of *Cercopithecus talapoin* (12.9 and 18.0).

These high deviations of individual values from the mean are obviously due to true intraspecific variation and not to difficulties in delineation, since the means for each of the various structures (except for total telencephalon and neocortex) are similar. These means, taken from all 18 species, vary from 3.2 to 5.0 as compared to the overall mean of 3.7 already mentioned, independent of difficulties in delineation.

Structures difficult to delineate (e.g. medulla oblongata, mesencephalon, dien-cephalon), have the same or even a lower average deviation than structures easy to delineate (e.g. cerebellum, hippocampus, olfactory bulb). The olfactory bulb, as a structure which can be delineated very easily and clearly, is represented 3 times in the figures higher than 10, thus pointing to a great variability. Less variability is found in the telencephalon as a whole and in the neocortex. The means are 1.4 and 2.8 respectively, thus lying below the above mentioned range (3.2-5.0). In general, the variation of the neocortex is very similar to that of the total telencephalon. The figure 2.8 is influenced by the very high value of *Talpa europaea* (21.3), which is the highest of all values under consideration. If we exclude this value, the average for the neocortex is 1.7. The more than 10 times higher deviation of *Talpa* seems to be due to an error and requires reconsideration.

All volume data smaller than 100 are indicated with one decimal fraction to the right of the unit figure, all data larger than 100 only in unit figures. In cases of very large figures, the last places to the left of the unit figure are of course fairly unimportant since a variability of up to 10% may be expected as above indicated.

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